

Statistics Notes

Statistics is the science of collecting, organizing, analyzing, interpreting, and presenting data.

Types of Data

Qualitative data describes categories or qualities. Quantitative data represents numerical values. Quantitative data can be discrete or continuous.

Population and Sample

A population is the complete group being studied. A sample is a smaller subset selected from the population.

Measures of Central Tendency

The mean is the arithmetic average. The median is the middle value after ordering observations. The mode is the most frequently occurring value.

Measures of Dispersion

Range is the difference between the maximum and minimum values. Variance measures the average squared deviation from the mean. Standard deviation is the square root of variance.

Probability

Probability measures the likelihood of an event and ranges from 0 to 1. An impossible event has probability 0, while a certain event has probability 1.

Correlation

Correlation describes the strength and direction of a relationship between two variables. Positive correlation means both tend to move in the same direction, while negative correlation means they tend to move in opposite directions.